

Quartz resonator

X26 (2026) — English translation

Error estimation is not required in this task.

Equipment: AC generator Oscilloscope Breadboard Set of jumpers 3 wires BNC-BNC Adapter “BNC-board” Package with electronic components, which includes: 9 capacitors; quartz resonator, the resonant frequency of which is about 2 MHz ; resistor $R_0 = 5.1\text{ k}\Omega$

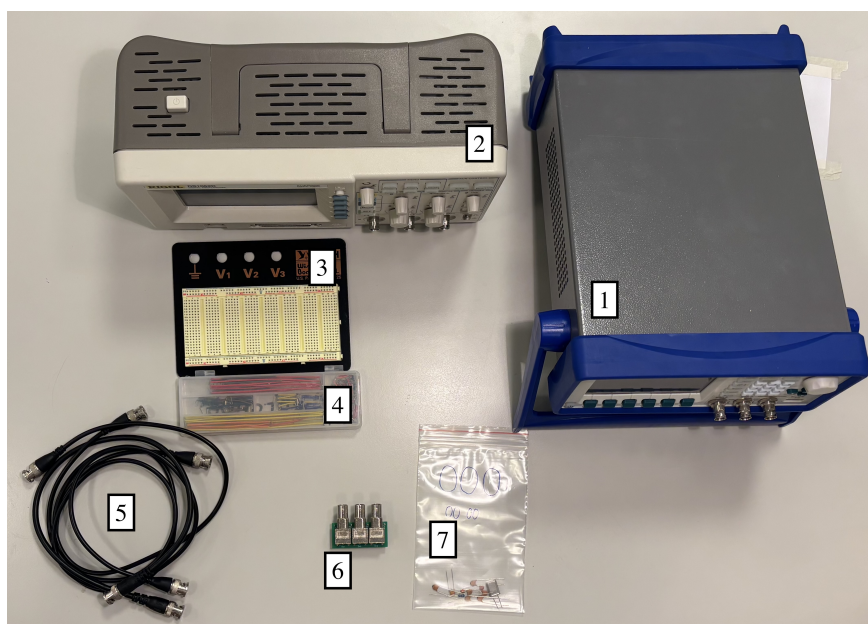


Figure 1: Figure 1.

Capacitor marking Capacity, pF 6 6.0 10 10 15 15 27 27 33 33 47 47 56J 56 68 68 82 82

A0 Write down the code for the package containing the electronic components on your answer sheets. Check that it is complete. **0**

A quartz resonator is an electronic component that uses the piezoelectric properties of quartz to generate electrical oscillations with high precision and stability. Based on its behavior in electrical circuits, a quartz resonator can, to a first approximation, be represented in the form of an equivalent electrical circuit shown in the figure. In this problem we will study a quartz resonator as an oscillatory circuit and find its parameters. It is known that the resonant frequency of a quartz resonator is about 2 MHz .

Note. Do not set the double voltage amplitude on the generator $2U_0 > 500\text{ mV}$. Otherwise, the operation of the quartz resonator in the same mode as at lower amplitudes is not guaranteed.

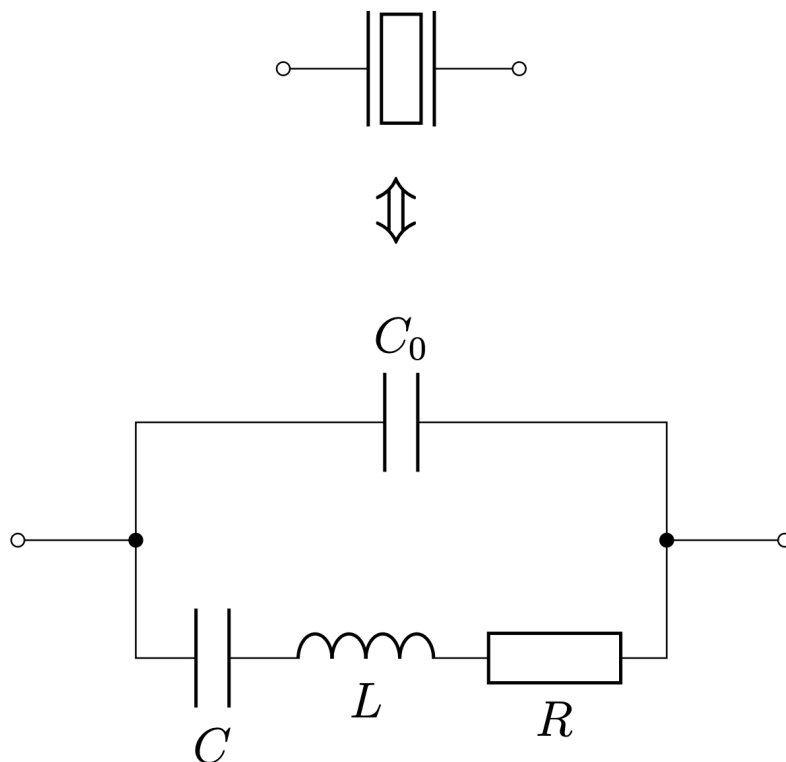


Figure 2: Figure 2.

For measurements, use the following scheme:

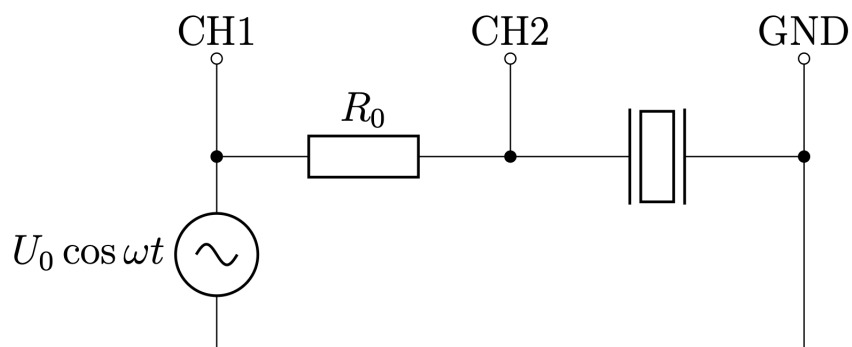


Figure 3: Figure 3.

A1	Find the signal frequency $f_{\max}$ at which the voltage amplitude across the quartz resonator is maximum. The difference between $f_{\max}$ and 2 MHz is no more than 1 kHz .	0.20
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A2	Record the ratio of the voltage amplitudes on the two channels of the oscilloscope $\frac{U_{\text{CH2}}}{U_{\text{CH1}}}$ at the signal frequency $f_{\max}$.	0.05
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A3	Remove the dependence of $2U_{\text{CH2}}$ and φ on the frequency f in the range $[f_{\text{max}} - 250 \text{ Hz}; f_{\text{max}} + 250 \text{ Hz}]$, where φ is the phase difference between the signals from channels 2 and 1. Measure at least 30 points from the given frequency range.	2.80
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A4	Plot a graph of the quartz resonator impedance modulus $ Z $ versus frequency f over the measured range.	2.50
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A5	From the plotted graph $ Z (f)$, determine the resonance frequency f_{res} (where $ Z $ is maximal), the antiresonance frequency f_{anti} (where $ Z $ is minimal), and the frequencies f_- and f_+ at which $ Z ^2$ is equal to the half-sum of the extreme values: $ Z ^2(f_{\pm}) = \frac{ Z _{\text{max}}^2 + Z _{\text{min}}^2}{2}$.	0.60
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We denote by f_{res} the resonance frequency and by f_{anti} the antiresonance frequency. The impedance of the quartz resonator can be written as

$$Z(\omega) = \frac{\omega_s^2 - \omega^2 + i\omega\alpha}{i\omega C_0 (\omega_p^2 - \omega^2 + i\omega\alpha)},$$

where ω is the angular frequency of the signal, and $\omega_s, \omega_p, \alpha$ do not depend on ω .

A6	Express $\omega_s, \omega_p, \alpha$ in terms of R, L, C, C_0 .	0.30
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Further in this problem, assume that near resonance the following approximations are valid:

$$Z(\omega) \approx \frac{\omega_s^2 - \omega^2 + i\omega_p\alpha}{i\omega_p C_0 (\omega_p^2 - \omega^2 + i\omega_p\alpha)}, \quad \alpha \ll \omega_p - \omega_s.$$

A7	Using the approximations described above, express $ Z ^2(\omega)$ in terms of $\alpha, \omega, \omega_s, \omega_p, C_0$.	0.10
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A8	Using the approximations described above, express $f_{\pm}$ in terms of $f_{\text{res}}, f_{\text{anti}},$ and α .	0.30
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In addition, within these approximations the following relation holds:

$$\varphi(f_{\text{anti}}) = \varphi(f_{\text{res}}) = 0,$$

where $\varphi(f)$ is the phase difference between channels 2 and 1 as a function of frequency. If an external capacitor C_{ext} is connected in parallel with the resonator, the resonance frequency f_{res} shifts.

A9	Measure the dependence of the shifted resonance frequency f_{res} on the external capacitance C_{ext} for all available capacitors.	0.80
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A10	Plot a linearized graph for the dependence from A9 and determine C_0 from this graph.	1.20
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A11	From the results of points A6, A10, obtain the values of L, C, C_0 .	1.30
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A12 From the results of points A4, A6, A10, obtain the value R .

1.20

Source: <https://pho.rs/p/4788>